ASSIGNMENT - 10

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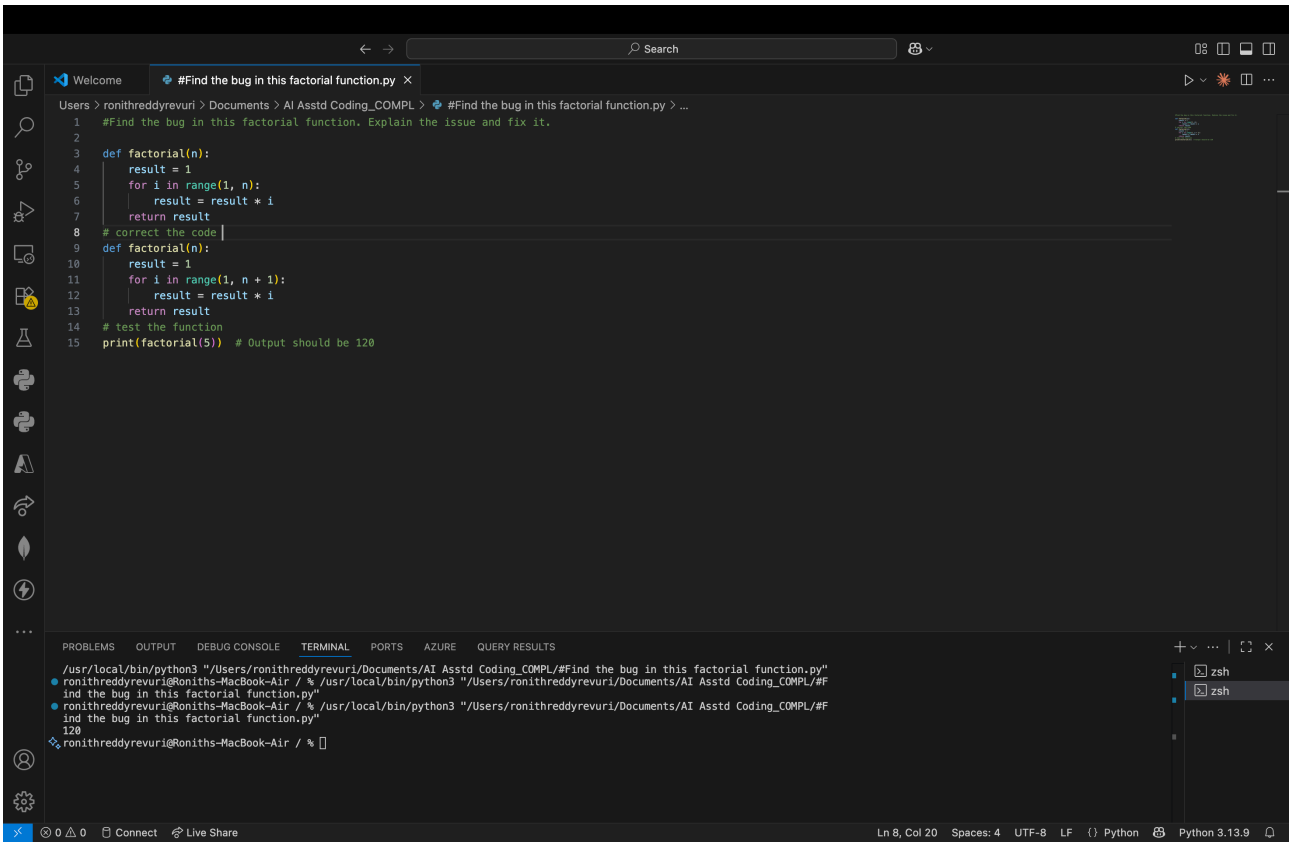
TASK - 1 :
AI-ASSISTED BUG DETECTION
SCENARIO: A JUNIOR DEVELOPER WROTE THE FOLLOWING PYTHON FUNCTION TO CALCULATE FACTORIALS:

PROMPT USED :
FIND THE BUG IN THIS FACTORIAL FUNCTION. EXPLAIN THE ISSUE AND FIX IT.

EXPLANATION :

THE REASON IS THE LOOP RUNS FROM 1 TO N−1 INSTEAD OF GOING TILL N.
BECAUSE OF THIS, THE LAST NUMBER (WHICH IS N) IS NOT INCLUDED IN MULTIPLICATION.



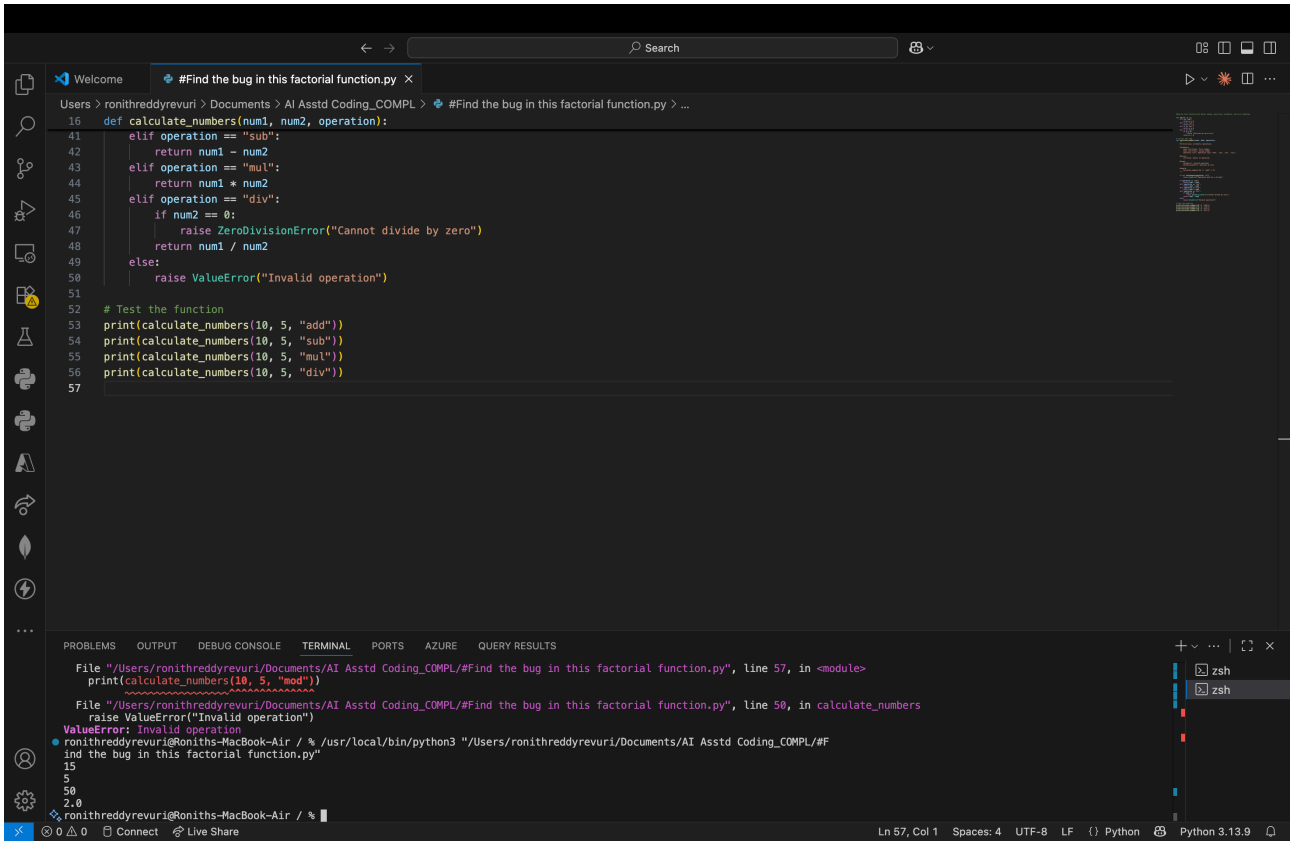
TASK 2 — IMPROVING READABILITY & DOCUMENTATION

PROMPT :
REWRITE THIS FUNCTION WITH BETTER NAMING, DOCSTRING, VALIDATION, AND ERROR HANDLING.

EXPLANATION :
THE IMPROVED VERSION:

- **USES MEANINGFUL NAMES**
- **INCLUDES A PROPER DOCSTRING**
- **HANDLES INVALID OPERATIONS**
- **ADDS DIVISION BY ZERO CHECK**
- **VALIDATES INPUT TYPES**

THIS MAKES THE FUNCTION EASIER TO UNDERSTAND AND SAFER TO US



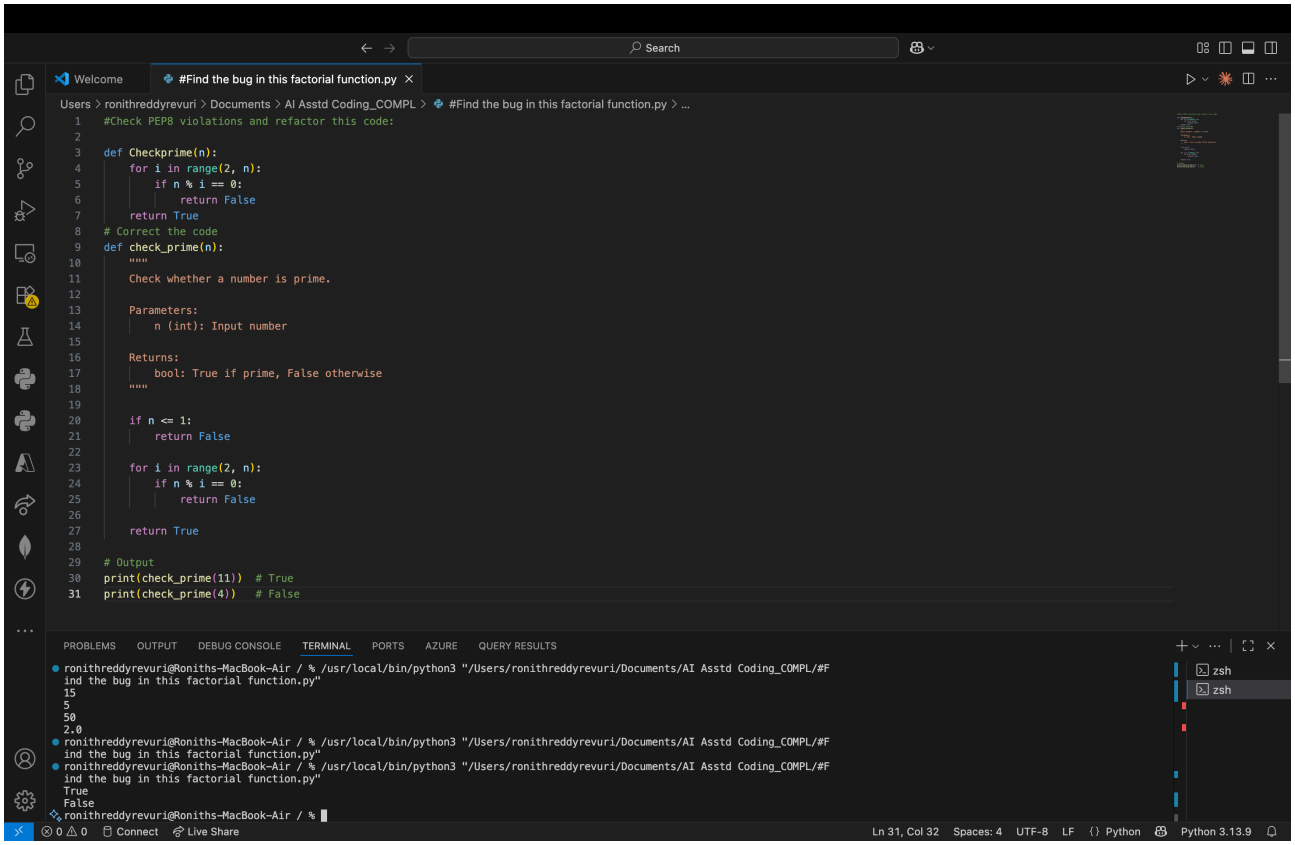
TASK-3 : ENFORCING CODING STANDARDS
SCENARIO: A TEAM PROJECT REQUIRES PEP8 COMPLIANCE. A DEVELOPER SUBMITS:

PROMPT:
CHECK PEP8 VIOLATIONS AND REFACTOR THIS CODE:

EXPLANATION :
THE CORRECTED VERSION:

- **FOLLOWS PROPER NAMING CONVENTIONS**
- **INCLUDES SPACING AND DOCSTRING**
- **HANDLES EDGE CASES**

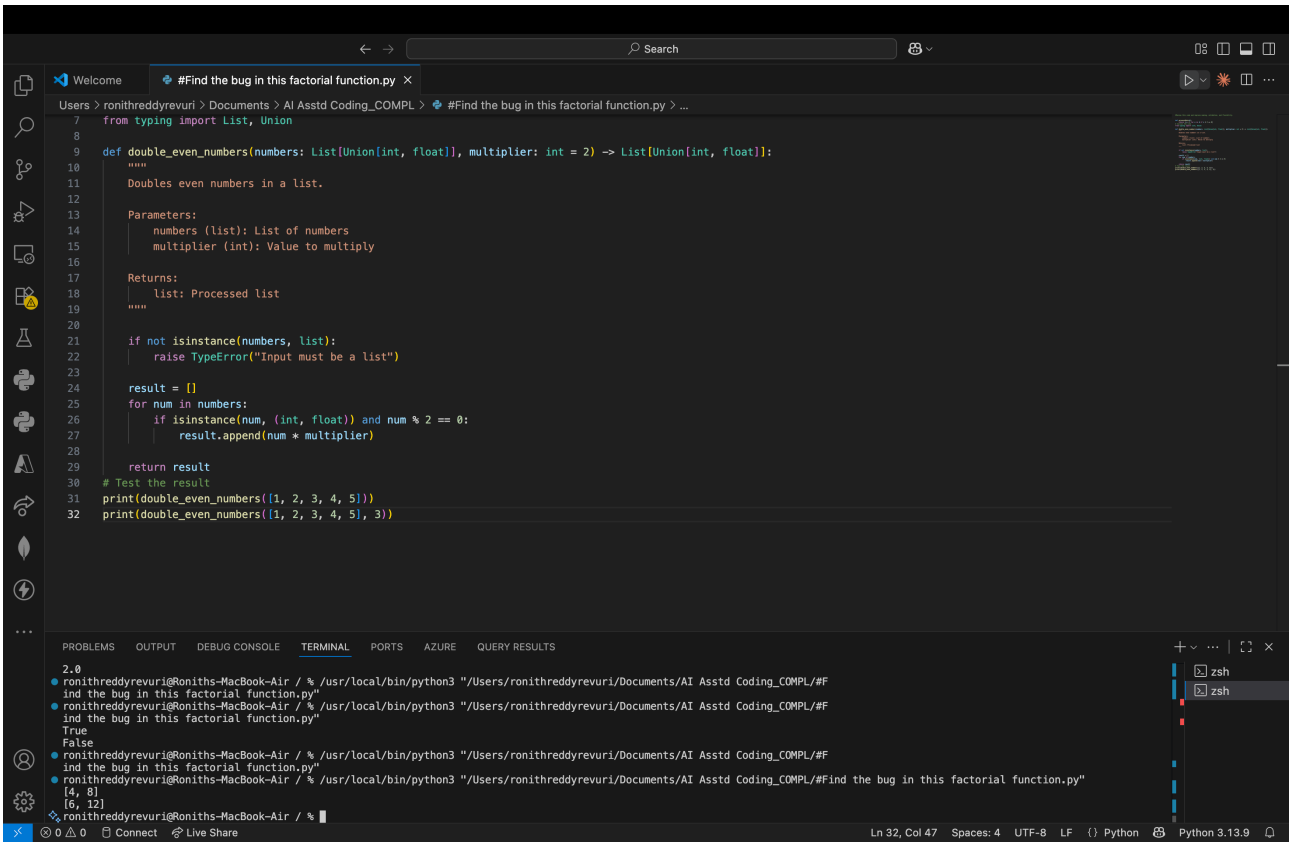
THIS IMPROVES CODE QUALITY AND CONSISTENCY.



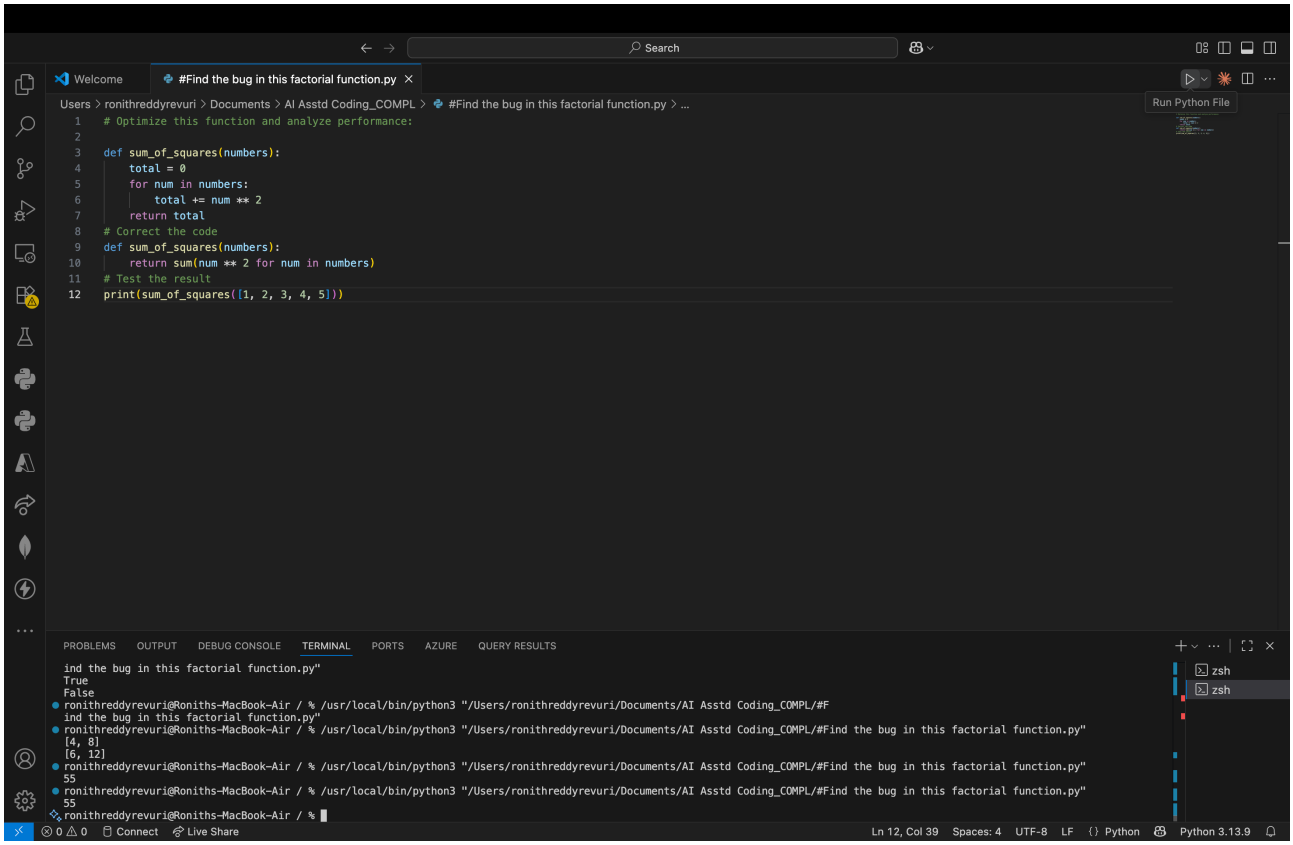
TASK-4 : AI AS A CODE REVIEWER IN REAL PROJECTS
SCENARIO:
IN A GITHUB PROJECT, A TEAMMATE SUBMITS:

PROMPT :
REVIEW THIS CODE AND IMPROVE NAMING, VALIDATION, AND FLEXIBILITY.

EXPLANATION :
THE IMPROVED VERSION IS MORE REUSABLE, FLEXIBLE, AND SAFER.
ALSO, I FEEL AI IS HELPFUL IN REVIEWING CODE, BUT IT SHOULD NOT REPLACE HUMANS COMPLETELY.
IT WORKS BEST AS A SUPPORT TOOL.



TASK - 5 : AI-ASSISTED PERFORMANCE OPTIMIZATION
SCENARIO: YOU ARE GIVEN A FUNCTION THAT PROCESSES A LIST OF INTEGERS, BUT IT RUNS SLOWLY ON LARGE DATASETS:

PROMPTS :
OPTIMIZE THIS FUNCTION AND ANALYZE PERFORMANCE:

EXPLANATION :
THE ORIGINAL FUNCTION USES A LOOP TO CALCULATE THE SUM OF SQUARES, WHICH WORKS CORRECTLY BUT IS SLIGHTLY SLOWER FOR LARGE INPUTS.